

Three-Loop Leading Singularities Summary

AntiGravity

ABSTRACT: This document summarizes the leading singularities of the 15 three-loop integrands. Integrands are classified by their set of leading singularities (and the number of such singularities). All failed or empty results are grouped into a single class.

Contents

1 Summary of Results 1

1 Summary of Results

Total integrands: 15. Total distinct classes: 8.

Class 1 (4 integrands)

Integrand Indices: 2, 3, 9, 11

Number of Leading Singularities: 1

$$\frac{1}{\sqrt{u^2 - 2uv - 2u + v^2 - 2v + 1}} \quad (1.1)$$

Class 2 (1 integrands)

Integrand Indices: 5

Number of Leading Singularities: 1

$$\frac{u}{(v-1)\sqrt{u^2 - 2uv - 2u + v^2 - 2v + 1}} \quad (1.2)$$

Class 3 (1 integrands)

Integrand Indices: 6

Number of Leading Singularities: 1

$$\frac{v}{u^2 - 2uv - 2u + v^2 - 2v + 1} \quad (1.3)$$

Class 4 (1 integrands)

Integrand Indices: 8

Number of Leading Singularities: 1

$$\frac{v}{(u^2 - 2uv - 2u + v^2 - 2v + 1)^{3/2}} \quad (1.4)$$

Class 5 (1 integrands)

Integrand Indices: 10

Number of Leading Singularities: 3

$$\frac{1}{\sqrt{u^2 - 2uv - 2u + v^2 - 2v + 1}} \quad (1.5)$$

$$\frac{1}{(v-1)\sqrt{u^2 - 2uv - 2u + v^2 - 2v + 1}} \quad (1.6)$$

$$\frac{v}{(v-1)\sqrt{u^2 - 2uv - 2u + v^2 - 2v + 1}} \quad (1.7)$$

Class 6 (1 integrands)

Integrand Indices: 12

Number of Leading Singularities: 2

$$\frac{v}{u^2 - 2uv - 2u + v^2 - 2v + 1} \quad (1.8)$$

$$\frac{v}{(v-1)\sqrt{u^2 - 2uv - 2u + v^2 - 2v + 1}} \quad (1.9)$$

Class 7 (1 integrands)

Integrand Indices: 13

Number of Leading Singularities: 2

$$\frac{1}{\sqrt{u^2 - 2uv - 2u + v^2 - 2v + 1}} \quad (1.10)$$

$$\frac{u}{(v-1)\sqrt{u^2 - 2uv - 2u + v^2 - 2v + 1}} \quad (1.11)$$

Class 8 (5 integrands)

Integrand Indices: 1, 4, 7, 14, 15

Result: Failed to evaluate / Timeout / Empty set.